PILOT POPULATION / DATASET GENERATION / FORECASTING

Multi-pilot training as dataset preparation

Project objective: forecast trained aircraft movement from one-sided observations

1 TRAIN PILOTS

Learn useful combat behaviour
Vary fixed reward priorities
Adapt to tactical opposition
Retain ten selected checkpoints

2 RECORD ENCOUNTERS

Freeze the learned pilot policies
300 encounters per pilot
3,000 episodes in total
Observe from the tactical aircraft

3 FORECAST MOTION

Use the recorded observations
Predict the trained pilot's motion
Position and velocity at +1 / 2 / 5 s
Assess on held-out scenarios

Multi-pilot RL is an upstream pre-processing and data-generation stage for forecasting.

Combat evaluations establish the quality of the flight-generating policies.

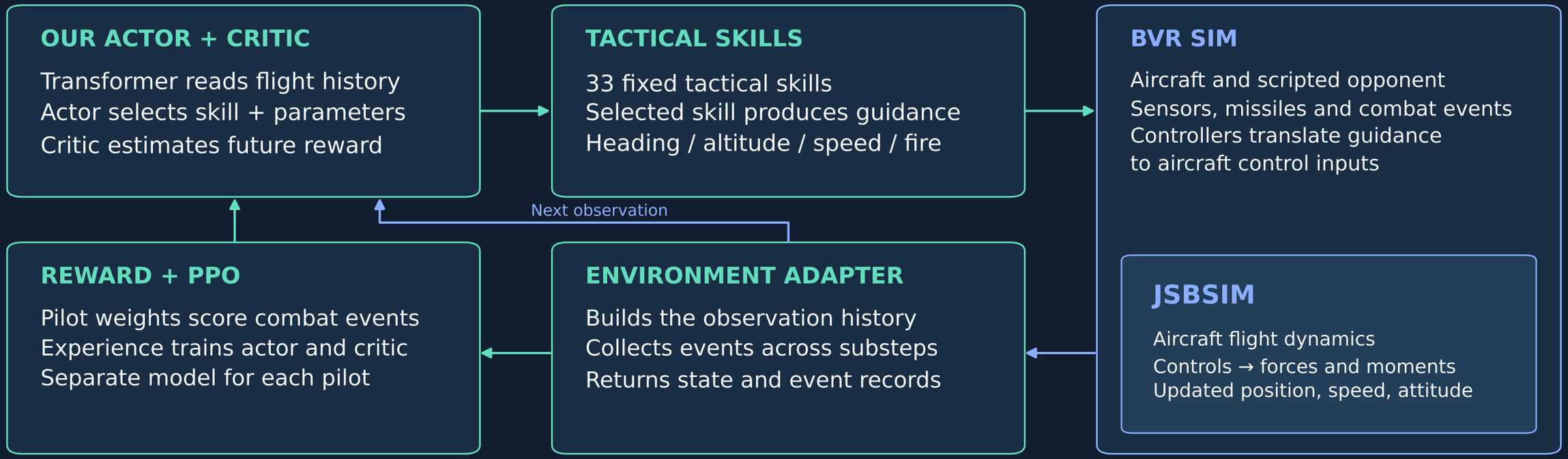
Different reward priorities provide behavioural variation for the observation dataset.

The final forecasting objective is learned from recorded movement; pilot rewards guide the earlier RL stage.

Saved experiment results | Scripted tactical observer | Forecasting final test remains reserved

Where RL, BVR Sim and JSBSim fit

The learned policy chooses behaviour; the simulator and flight model determine what happens



Current setup: F-16 versus F-16, 1-v-1 • scripted simple or tactical opponent • ten independent pilots

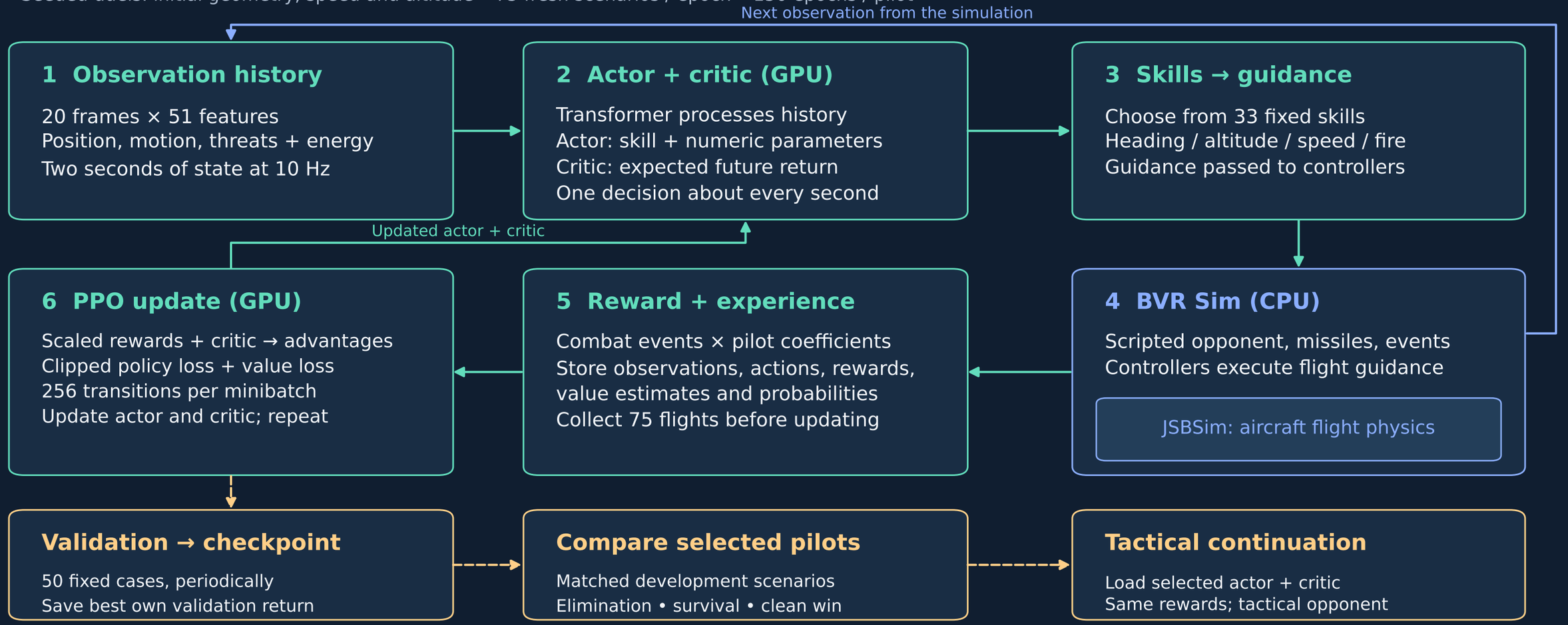
CPU workers advance independent duels. GPU batches policy inference and PPO updates.

Scenario setup initializes the encounter. JSBSim supplies aircraft physics; BVR Sim supplies the combat environment.

RL pipeline: decisions, simulation and learning

Ten independent pilots • individual reward profiles and checkpoints • shared scenario schedules

Seeded duels: initial geometry, speed and altitude • 75 fresh scenarios / epoch • 150 epochs / pilot

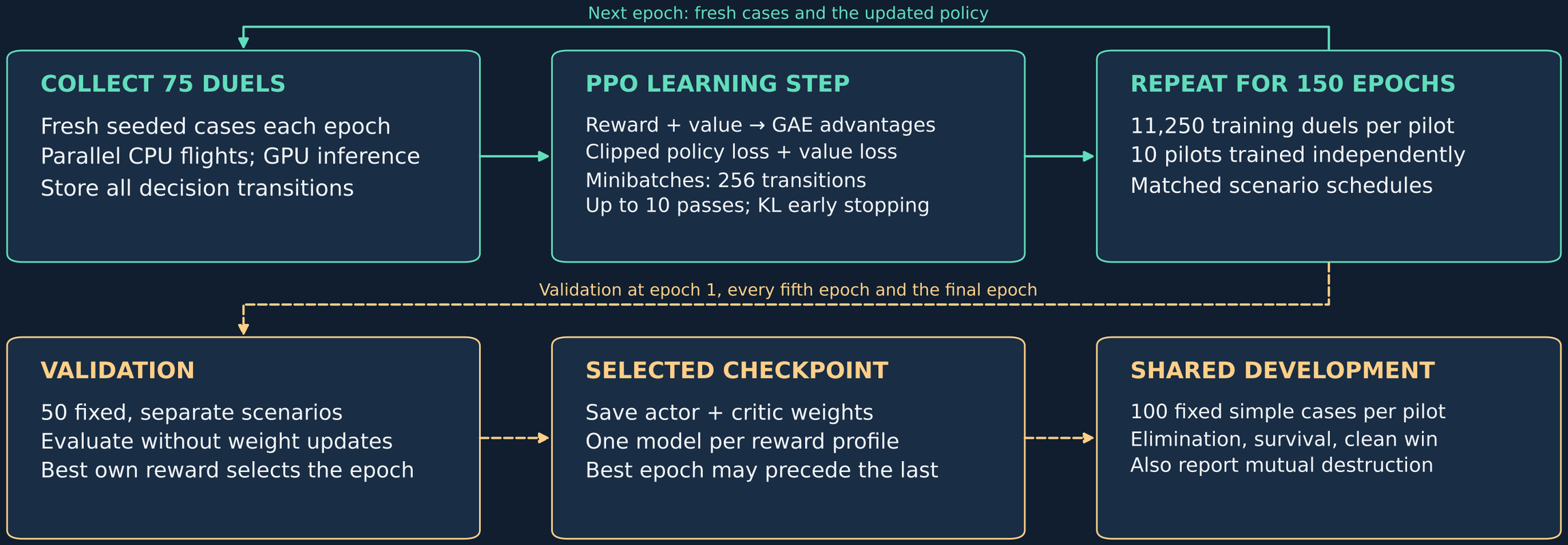


CPU workers parallelise flights within each pilot. Final testing uses untouched cases after development choices are frozen.

Saved experiment results | Scripted tactical observer | Forecasting final test remains reserved

Training, checkpoint selection and comparison

Training changes weights • validation selects checkpoints • development compares experiments



Tactical stage: preserve selected parents → load actor + critic → fresh Adam → 150 further epochs against tactical.

Validation and development cases have no training updates. Final testing waits until experimental choices are frozen.

Simple-opponent results

Shared development benchmark • 10 pilots × 100 matched cases • seed 211307

Population	Elimination	Survival	Clean win	Mutual loss
Original	59.8%	57.6%	28.2%	31.6%
Reduced learning rate	60.8%	66.5%	36.8%	24.0%
Selected-skill probabilities	72.2%	68.8%	50.0%	22.2%
Per-skill parameter heads	65.4%	68.3%	42.5%	22.9%

Lower learning rate: the main observed gain was survival, with similar elimination frequency.

Selected-skill probabilities: more eliminations, with a smaller further improvement in survival.

Elimination and survival overlap. Clean win records both occurring in the same flight.

Experimental development

Independent population training, controlled PPO experiments and tactical adaptation

04 → 05	One PPO pilot became an arbitrary-size population with separate rewards and checkpoints.
Reward diversity	Monotonic sweep → relative Halton profiles → seeded, independent uniform ranges.
Evaluation design	Recorded training/validation and a fixed benchmark; final testing after development.
Optimisation	Lower learning rate and selected-skill probabilities improved clean wins.
Bounded trials	More parameter heads underperformed; the terminal-bonus trial stopped incomplete.
Tactical	The stronger shared-head population was retained and adapted to scripted tactical.

Scenario collection and PPO updates

75 scenarios per epoch for each pilot; shared scenario schedules support matched comparisons

OBSERVE

20 frames × 51 features
2 seconds of flight history

→

CHOOSE

One of 33 skills
+ its numeric parameters

→

SIMULATE

Execute about 1 second
Record events and rewards

→

UPDATE

Estimate advantages
PPO minibatches of 256

Scenarios and minibatches

A scenario is one duel with seeded starting geometry, speeds and altitudes (up to 120 seconds).
75 scenarios produce many decision transitions. A PPO minibatch contains 256 transitions.
150 epochs × 75 scenarios = 11,250 training flights per pilot; 112,500 across the population.

Actor–critic network architecture

Shared transformer: width 64 • 4 attention heads • 2 layers • shared parameter predictor

ACTOR: policy distribution

Predict a distribution over 33 skills
and 16 numeric parameter slots.
Only parameters used by the chosen
skill count in its PPO probability.

CRITIC: value estimation

Predict future discounted reward.
Compare rewards + future value with
that prediction to estimate advantage.
Positive advantage favours the action.

$$\log \pi(a \mid s) = \log \pi(k \mid s) + \sum_{j \in \text{active}(k)} \log p(z_j \mid s)$$

The selected-skill correction changed the learning objective, without adding network capacity.

Reward profiles and common evaluation metrics

Each pilot draws its coefficients once using a reproducible random seed (42)

Crash / being shot down	Each independently uniform: −450 to −300
Lock / launch with lock	Each independently uniform: 0 to 2
Avoid an incoming missile	Uniform: 0 to 15
Launch without lock	Uniform: −12 to −4
Opponent destroyed / clearance	Fixed: +250 / +0.02

Common outcome scoring

Clean win +100 • survive without elimination 0 • mutual destruction −50 • own loss −100

Interpretation of simple-opponent results

Changes on the fixed development benchmark • percentage points relative to the matching baseline

LOWER LEARNING RATE

Eliminations: +1.0 points.
Survival: +8.9 points.
Clean wins: +8.6 points.
Mutual destruction: −7.6 points.

SELECTED-SKILL PROBABILITIES

Eliminations: +11.4 points.
Survival: +2.3 points.
Clean wins: +13.2 points.
Mutual destruction: −1.8 points.

Bounded trials

Per-skill heads: eliminations fell 72.2% → 65.4%; survival was similar, 68.8% → 68.3%.
Bonus trial (6 matched pilots): eliminations 72.8% → 72.2%; survival 68.3% → 63.8%.
Both scored fewer clean wins; the shared-head selected-skill population was retained.

Recorded training and validation

Population mean own return • identical reward profiles across runs • 150 epochs per pilot



Own return describes learning. Assess combat using elimination, survival, clean wins and mutual destruction.

Tactical adaptation: experimental design

Same reward profiles, skills, physics and PPO settings; new scripted opponent behaviour

BEFORE

Freeze the 10 selected-skill parents.
Assess all on 300 tactical cases.
Keep the originals unchanged.

FINE-TUNE

Load each pilot's actor + critic.
Fresh Adam; LR 0.00003.
150 epochs × 75 fresh tactical flights.

SELECT

50 separate validation cases.
Own reward selects the checkpoint;
initial weights are also eligible.

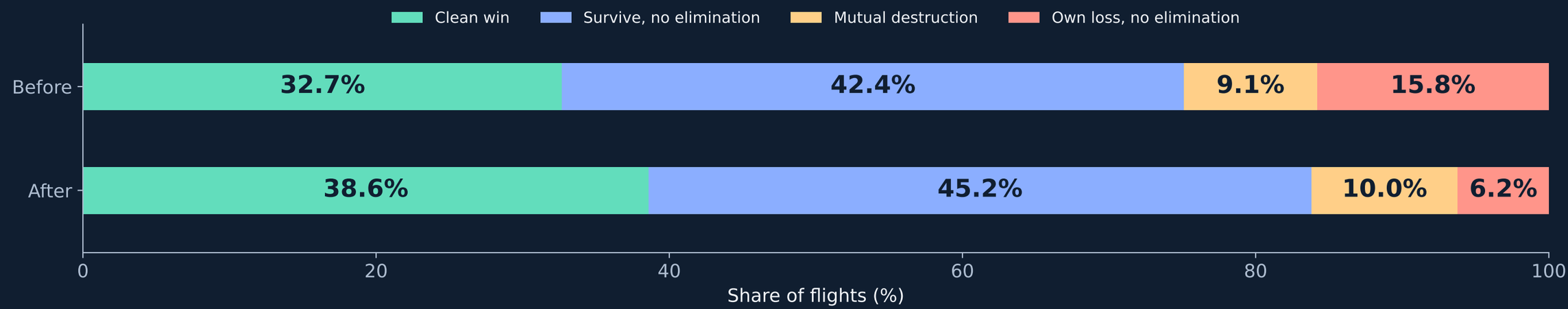
COMPARE

Repeat the same 300-case assessment.
Use a separate 100-case confirmation
for both parent and adapted pilots.

Tactical adaptation outcomes

Historical 100-case confirmation • seed 1220000 • the same cases before and after adaptation

Population	Elimination	Survival	Clean win	Mutual loss
Before	41.8%	75.1%	32.7%	9.1%
After	48.6%	83.8%	38.6%	10.0%



Clean-win gain: +5.9 pp | Paired scenario bootstrap 95% interval: +1.2 to +11.0 pp

Per-pilot confirmation results

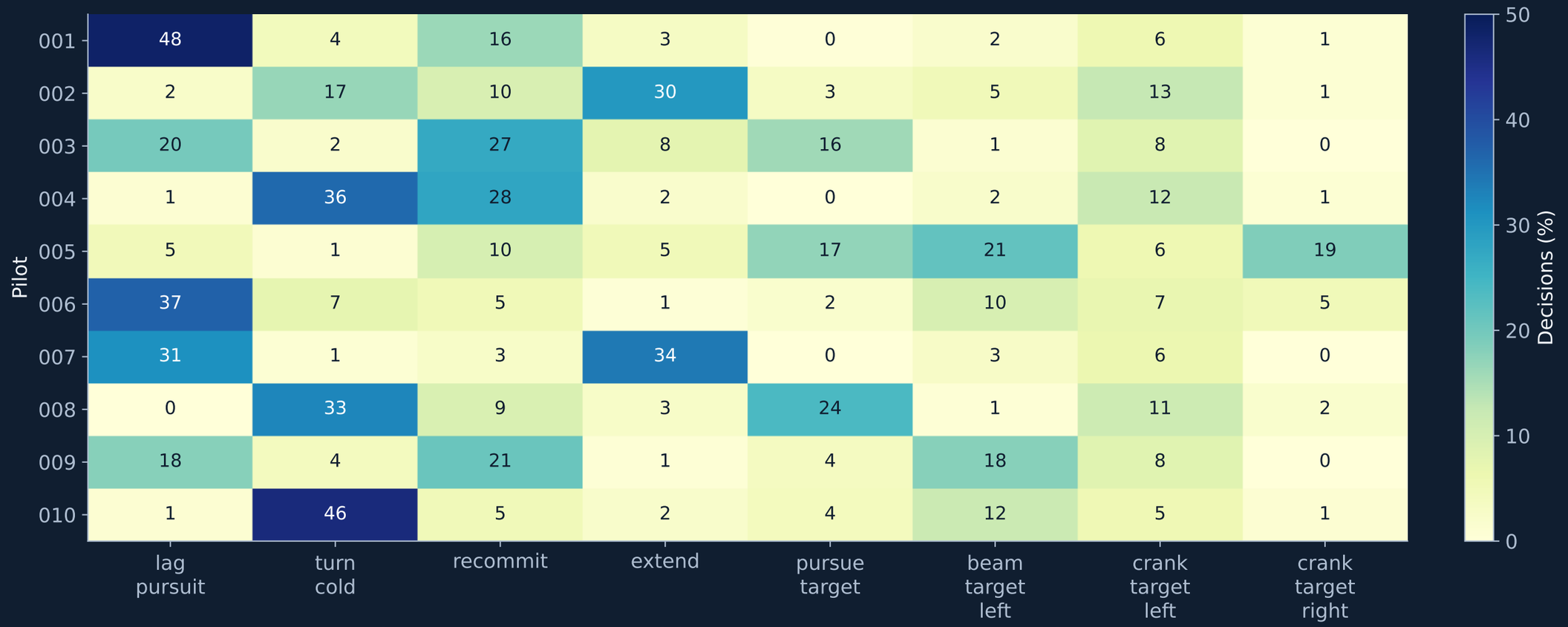
Nine clean-win rates improved and one tied on the same 100 tactical confirmation cases

Pilot	Elimination	Survival	Clean win	Mutual loss
001	42% → 45%	72% → 81%	34% → 34%	8% → 11%
002	45% → 49%	75% → 84%	32% → 38%	13% → 11%
003	41% → 46%	74% → 85%	32% → 37%	9% → 9%
004	42% → 49%	79% → 84%	33% → 39%	9% → 10%
005	38% → 52%	74% → 86%	31% → 44%	7% → 8%
006	37% → 49%	65% → 83%	30% → 39%	7% → 10%
007	43% → 51%	80% → 85%	33% → 43%	10% → 8%
008	51% → 50%	76% → 83%	37% → 38%	14% → 12%
009	41% → 44%	76% → 85%	34% → 35%	7% → 9%
010	38% → 51%	80% → 82%	31% → 39%	7% → 12%

005 after adaptation: 52% eliminations, 86% survival, 44% clean wins. Small gaps remain uncertain.

Behavioural variation after adaptation

Observed decision frequencies on confirmation flights • descriptive measures, not aggression rankings



001 favours lag pursuit; 010 often turns cold; 005 mixes beaming, cranking and pursuit.

Generalisation and evaluation limitations

Matched performance improvements and the scope of the available evidence

OBSERVED RESULTS

100-case confirmation: +5.9 pp clean wins.
300-case assessment: +6.9 pp clean wins.
Survival improved in both suites.
All 10 training runs completed; parents and frozen source files verified unchanged.

EVALUATION LIMITATIONS

One training seed per reward profile.
No post-adaptation test against simple.
Mutual destruction did not improve on the confirmation suite.
No universal best pilot or style proven.

No direct seed overlap was found across training, selection and tactical report suites.

Held-out gains do not establish the absence of overfitting to this simulator or opponent.

From trained pilots to forecasting data

Observer: scripted tactical F-16 (B01) • prediction target: trained pilot F-16 (A01)

3,000 RECORDED ENCOUNTERS

300 matched starting scenarios × 10 frozen pilot policies
Five seconds of past observations → movement at +1, +2 and +5 seconds
Each horizon predicts a 3D position and a 3D velocity vector

Partition	Scenario groups	Episodes
Training	210	2,100
Validation	45	450
Reserved test	45	450

TWO SHARED FORECASTERS

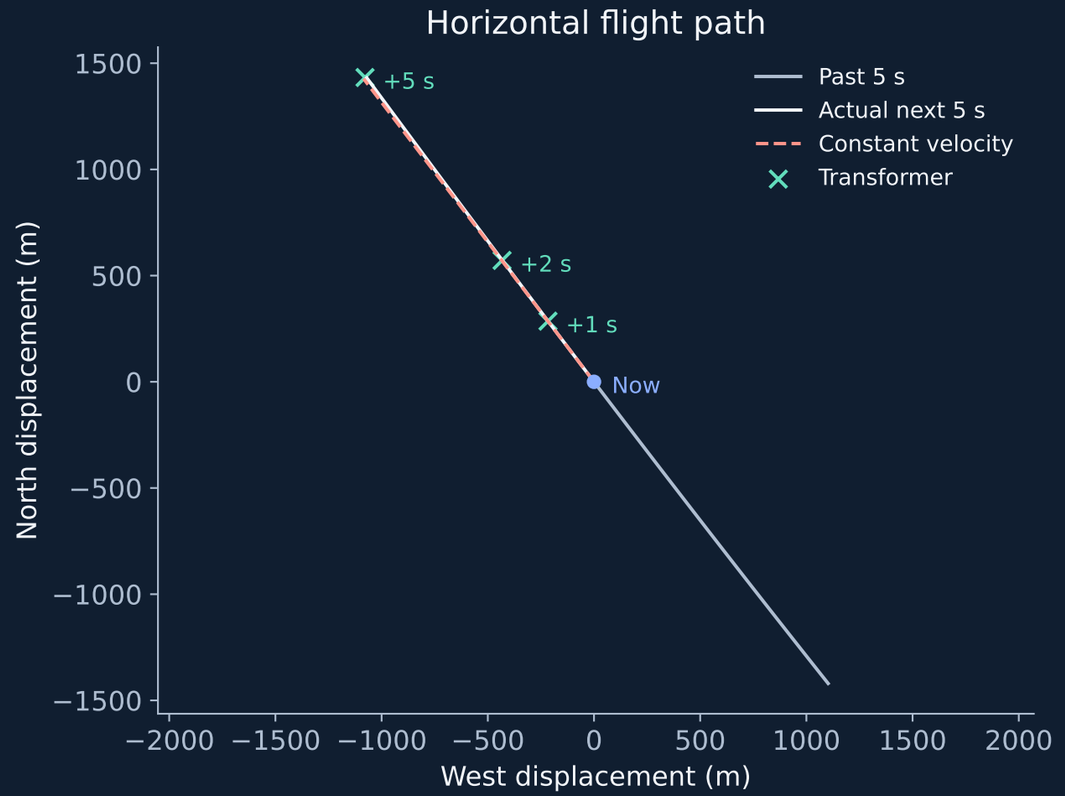
Transformer and GRU
Each learns across all ten pilots
Pilot identity is excluded
Future states supply training targets

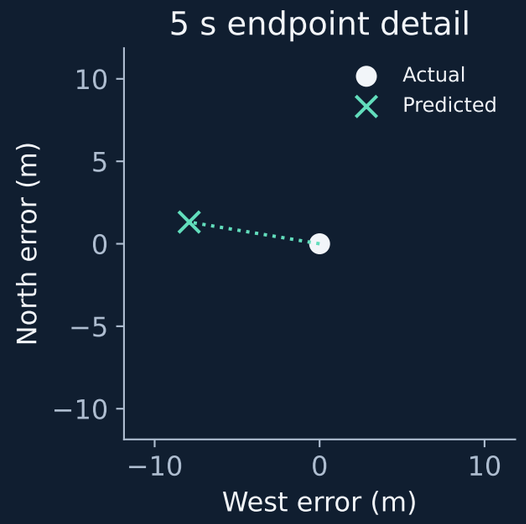
Whole scenario groups stay together across all ten pilots when splitting the dataset.

Forecast training reuses recorded flights; it does not rerun the simulator or update the pilots.

Location forecast and prediction error

One saved validation example • observed history ends at the forecast origin





8.1 m

3D position error in this example

1.0 m

Vertical component (abs. height error)

Position error = distance from predicted to actual location.

Velocity error = difference in movement speed and direction.

Pilot 004 • forecast at 71 s • example chosen at the median 5 s error

The enlarged view shows horizontal error; the reported distance also includes height error.

Initial forecasting performance in physical units

450 validation encounters • mean 3D error with equal episode weighting • lower is better

POSITION ERROR (METRES)

Method	+1 s	+2 s	+5 s
Constant velocity	9.15	36.18	221.46
Constant acceleration	0.87	5.23	63.94
GRU	0.44	1.93	20.65
Transformer	0.48	2.02	19.09

VELOCITY ERROR (METRES / SECOND)

Method	+1 s	+2 s	+5 s
Constant velocity	17.91	35.71	86.97
Constant acceleration	2.18	7.03	35.64
GRU	0.91	2.40	12.11
Transformer	0.98	2.37	10.94

At 5 s: Transformer reduces mean location error by 70% versus constant acceleration.

HOW OFTEN IS THE TRANSFORMER CLOSE TO THE ACTUAL LOCATION?

89.5%

within 1 m at +1 s

92.1%

within 5 m at +2 s

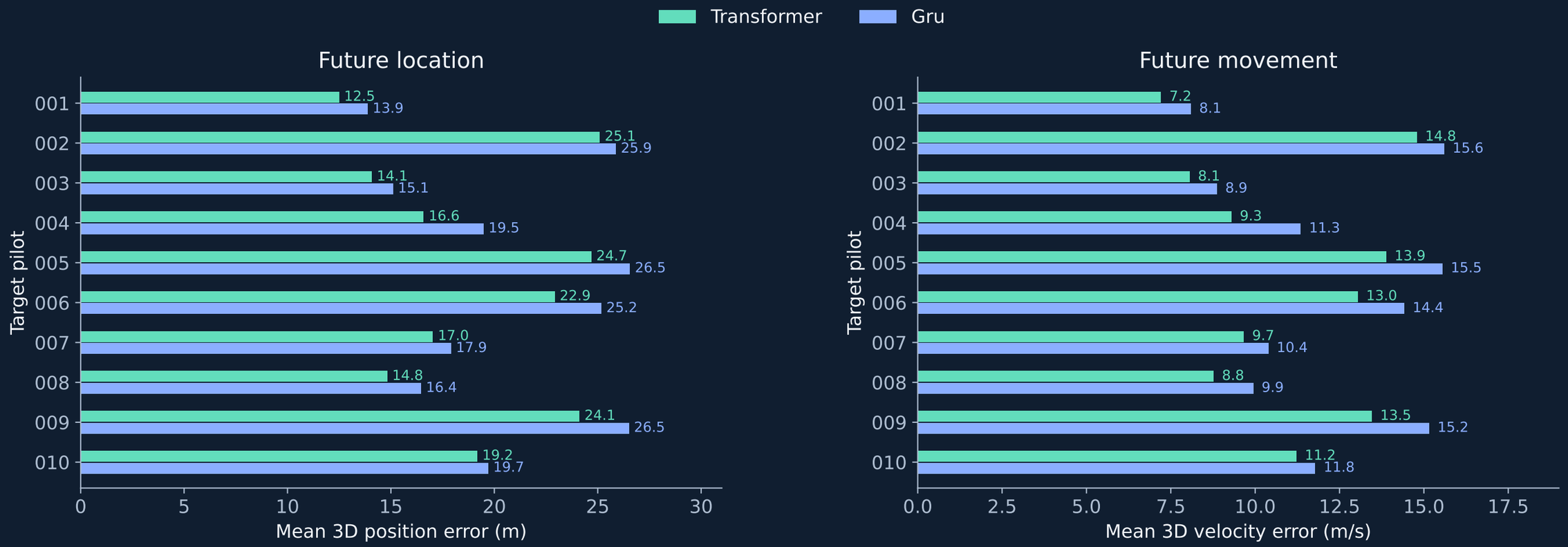
80.0%

within 25 m at +5 s

These are measured tolerance rates, not guarantees. Final test scenarios remain unevaluated.

Forecasting difficulty across the ten pilots

Same 45 held-out starting scenarios per pilot • five-second horizon • lower is better



Pilot 001 is easiest to predict. Pilots 002, 005, 006 and 009 have larger five-second errors.

Transformer improves both five-second errors for all ten pilots relative to GRU.

Predictability is a different property from combat effectiveness or aggression.

From motion forecasts to interpretable manoeuvres

Current outputs describe physical movement; behavioural labels are a subsequent interpretation layer

CURRENT CAPABILITY

Predict future position and velocity
Assess errors in metres and m/s
Compare error across horizons and pilots
Display actual and predicted trajectories

PROPOSED INTERPRETATION

Turn direction from horizontal velocity
Climb / descend from vertical motion
Acceleration from change in speed
Evaluate these labels against actual motion

A named catalogue skill expresses intent; several skills can produce similar observed motion.

Predicting one of the 33 internal skills would be a separate supervised classification task.

Evidence scope: known pilot population • idealised tracks • one split and training seed

Next validation questions: sensor noise, unseen pilots, difficult manoeuvres and the reserved test set.

Portable reports and supporting data

PDF and self-contained HTML formats; no notebook runtime, Python, GPU or LFS download required

REPORTS

presentation.pdf — slide deck
presentation.html — offline browser deck
README.md — technical report

DATA

data/results.json — pilot results
data/forecasting/metrics.csv — forecast errors
data/forecasting/summary.json — provenance

Principal findings

Population training and tactical adaptation prepared ten flight-generating policies.
Their recorded encounters support forecasts from the tactical observer's perspective.
Initial neural forecasts improve on constant-motion references on held-out scenarios.